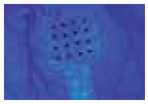
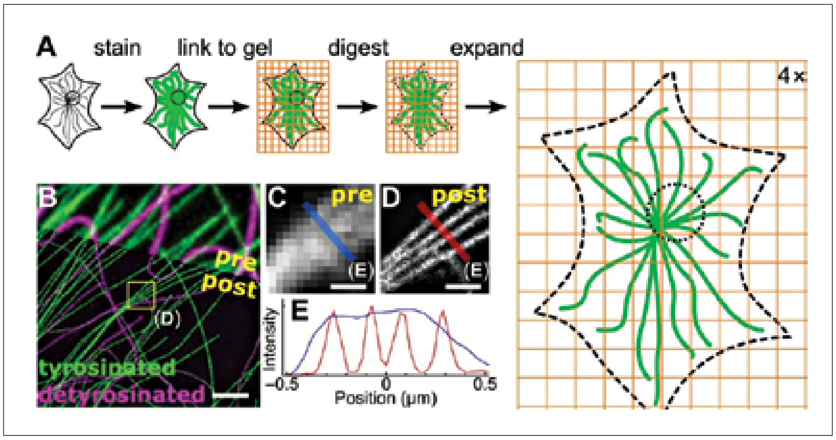
Feather could be better than Velcro
The zipping mechanism of bird feathers could provide a model for new adhesives and new aerospace materials. **https://dgit.in/feb19-83**



Graphene to hear brain whispers
Newly developed graphene-based implant can record brain activity at below 0.1 Hz and over large areas. **https://dgit.in/feb19-84**



How expansion microscopy works

as the hydrogel scaffold material. This scaffold is then immersed in a fluorescein molecule solution. These molecules, when and where they're activated with light, attach to the scaffold and act as anchors for other types of molecules that the researchers might add eventually.

"It's a bit like film photography — a latent image is formed by exposing a sensitive material in a gel to light. Then, you can develop that latent image into a real image by attaching another material, silver, afterwards. In this way implosion fabrication can create all sorts of structures, including gradients, unconnected structures, and multimaterial patterns," says Daniel Oran, one of the two lead authors on the paper. The other senior author is Adam Marblestone, a Media Lab research affiliate.

Now comes the actual shrinking part. Once the desired molecules have been attached to the intended locations, researchers add an acid to shrink the entire structure. The negative charges in the polyacrylate gel are blocked by the acid, removing their ability to repel each other which causes the gel to collapse and contract unto itself. This allows a 10-fold reduction in each dimension, resulting in a 1000-fold overall reduction. Overall, this process allows for better resolution than pre-existing methods, something that the researchers plan to improve even further. This process could democratise nanotechnology and provide scope for applications in robotics, optics, and many other fields. What this cannot do is build you an Ant-Man suit. Why not? Let's get into the physics of that and see how many leaps of faith it takes us to rationalise the possibility of Pym-particles.

PYM-POSSIBLE

Back in 2012, the verification of the long-predicted Higgs Boson answered a fundamental question of physics: why do different particles, such as electrons or protons, have different masses. The Higgs field can be thought of as an omnipresent fluid for all particles to move through. The viscosity of this fluid depends on the strength of the particle's interactions with the Higgs field and gives the particle its mass. In the fictional Marvel universe, Dr Hank Pym had discovered something similar half a decade earlier.

The Pym particle, Hank Pym's discovery that possesses, or rather grants, the ability to shrink and grow, can be thought of as a consequence of a Pym field around us.

Source: Marvel Cinematic Universe

Pym Particles

It doesn't need to be explained that this technology is quite impossible right now. However, it is interesting to explore why. Take the example of a person - it is impossible to make yourself larger or smaller. If you were to try and do it by modifying the number of atoms that make you up, there are a number of problems - Where would you store the removed atoms? Where would the additional atoms come from? How would you know how to arrange them? On the other hand, you could ponder about squeezing them closer together. Except for the fact that they are already in physical contact and the repulsive forces between electrons from neighbouring atoms are enough to keep them from getting any closer without the application of pressure that is typically found at places like the centre of the Earth.

What you could try to do, somehow, is alter the constants that determine the size of atoms in your body. For instance, if Planck's constant were to become 10 times smaller, the resulting average radius of all atoms affected would be reduced one hundred times. This would shrink a 6-foot tall person to 3/4th of an inch, small enough to be able to ride a carpenter ant. However, they'd retain the same mass, compressed into a very tiny volume, leading them to fall right through the floor. On the other hand, if the mass is also reduced by the same factor as the size, the density would remain constant and they wouldn't fall through the floor. However, this leads to another problem - how would this Ant-Man pack a punch? The only way this is possible is if Ant-Man can willingly couple and decouple the Higgs Field and the Pym Field so that he can become his full weight at ant-size when he is packing a punch, and remain lightweight enough to ride a flying ant otherwise. How does he do that? Maybe check with him once he is out of the Quantum Realm in May.